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Enhancement of blowout limit in a Mach 2.92 cavity-based scramjet combustor by a gliding arc discharge

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Abstract

Plasma-assisted combustion (PAC) has shown excellent performance in the ignition and combustion enhancement of scramjet combustors. A multi-channel gliding arc (MCGA) plasma placed on the sidewall surface of a scramjet combustor was employed to enhance combustion near the flame blowout limit (BL) in a C_2H_4-fueled and cavity-based model scramjet combustor. Wall static pressure measurements, simultaneous 10 kHz CH^* chemiluminescence imaging from the top and side views, optical emission spectroscopy, and discharge waveform measurements were used to elucidate the combustion physics throughout the PAC events. For the current cavity configuration and fueling condition, a general estimate of the flame BL in a rich-fuel environment was provided, in which a fuel flow rate (FFR) was 7.5 g/s. Once the FFR was exceeded, the flame was unstable and blown out, whereas the MCGA plasma demonstrated an excellent flame stabilization ability. The flame BL of the scramjet combustor in the presence of the MCGA plasma has an increase of ~29%. Two PAC processes, including enhancement mode and re-ignition mode, were distinguished to play a vital role in extending the flame BL. The two PAC processes showed that the spark-type discharge generated by the MCGA could ignite the local fuel-rich mixture and increase the flame BL. The local ignition is dominated by the continuous accumulation of the massive flame kernels with the local fuel-rich mixture, as well as more heat and species exchanges.

Keywords: Gliding arc discharge; Scramjet combustor; Plasma-assisted combustion; Blowout limit; Cavity flameholding

1. Introduction

Scramjet engines [1] are considered one of the most promising vehicles for future air-breathing hypersonic flight. Flame stabilization is crucial to the efficient, safe operation of scramjet engines. Cavi-

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ties [2] are generally used as flame holders in scramjets due to the low-speed recirculation region and less total pressure loss. However, the cavity flameholder has a flame blowout limit (BL) [3], which is dependent on cavity configurations [4], fueling schemes [5], and inflow conditions [6]. The flame BL for a specific cavity configuration is commonly evaluated by fuel flow rates (FFRs) seeded into scramjet combustors. Once the flame BL is reached in a highly fuel-rich or fuel-lean environment, the flame cannot be stabilized, so enhancing the flame BL of scramjet combustors is essential.

Many previous studies mainly focused on evaluations of the flame BL in scramjet combustors using theoretical analysis [3], numerical simulation [7], and experimental observation [6]. Passive strategies, for example, the promotion of cavity configurations and fueling schemes, are proposed to extend the flame BL, which is complex to use and restricted in few inflow conditions. In this case, developing active strategies, for example, plasma-assisted combustion (PAC), to enhance the flame BL is vital [8, 9].

PAC has been considered an innovative method for enhancing ignition, flame propagation, and flame stabilization through thermal, chemical, and transport effects in the past decades. Do et al. [10] achieved ignition enhancement by a nanosecond pulsed high-voltage discharge, whereas the static temperature was higher than the auto-ignition temperature of H_2. Matsubara et al. [11] succeeded in combustion enhancement in a step-based combustor using a plasma torch and a dielectric barrier discharge (DBD), which combined the characteristics of equilibrium plasma and non-equilibrium plasma. The long-lifetime O_3 generated by the DBD plasma showed kinetic effects on combustion enhancement. Gliding arc plasmas [12] could generate spark-type and glow-type discharges, which exhibit both characteristics of equilibrium plasma and non-equilibrium plasma, respectively. In our previous work, a multi-channel gliding arc (MCGA) plasma was employed in a cavity-based combustor to achieve successful ignition [13], combustion enhancement [14], and oscillation suppression [15] due to the heat release and reactive species generated by the MCGA plasma.

Although the MCGA plasma was widely used to enhance ignition and combustion in scramjets, the research about extending the flame BL by plasmas is still rarely reported. Few studies depict the dynamic processes of flame stabilization, and the mechanism of plasma-enhanced BL in scramjets remains unclear. The present work aims to fill the gap by investigating the effects of MCGA plasma on the enhancement of the flame BL in scramjet combustors. To help achieve the above objective, an MCGA plasma arranged on the sidewall of a Mach 2.92 scramjet combustor was designed. Dual-view high-speed photography and static pressure measurements were simultaneously utilized to analyze the processes of the plasma-enhanced BL. Voltage and current measurements and optical emission spectroscopy were employed to elucidate the mechanism. The investigation can provide a comprehensive understanding of how the MCGA plasma enhances the flame BL in a model scramjet combustor.

2. Experimental description

2.1. Scramjet combustor

The experiments were conducted in a direct-connect test facility, which was described in detail in [16–18]. The facility consisted of a vitiated air supply system, a nozzle designed for Mach 2.92, a 40 × 50 mm cross-section isolator, a supersonic model combustor, and an expansion section. The high-enthalpy vitiated air flow with a 21% O_2 mole fraction was supplied by burning absolute ethanol (99.9% purity) and oxygen in an air heater. The supersonic inflow had a total pressure of 2.35 MPa, and a total temperature of 1590 K. A total mass flow rate of the supersonic gas flow was 1 kg/s.

A schematic of the model scramjet combustor with a cavity flame holder is shown in Fig. 1. The supersonic inflow passed through the isolator and entered the combustor. The 20 × 90 mm with a 45° aft-ramp cavity flame holder was equipped with an expansion angle of 2°. The room-temperature C_2H_4 was injected from the aft ramp at the height of 5 mm above the floor. Four sonic C_2H_4 orifices with a diameter of 1 mm were arranged equidistantly with a distance of 10 mm on the aft ramp. The FFRs in the present work were 7.5–11.8 g/s. A pressure transducer was used to measure the wall pressure at 10 mm downstream of the trailing edge of the cavity. The sampling frequency and measurement uncertainty of the pressure transducer was 100 Hz and ±0.1%, respectively.

2.2. MCGA plasma

An MCGA plasma igniter was fixed on the sidewall of the combustor, as shown in Fig. 1. The main reason for placing the ignitor was to increase the interaction area between the MCGA plasma and the flame. The ignitor consisted of six tungsten-made high-voltage (HV) electrodes, a filled ceramic, and a stainless-steel case. The stainless-steel case and the combustor acted as ground electrodes. The HV electrodes were supplied by an HV alternating current power supplier (CG10000FG, Suman Technology, China). The shortest distance between the electrodes and the inner diameter of the stainless-steel case were 7 and 36 mm, respectively. The MCGA had negligible effects on blocking the flow path because the blockage ratio of the MCGA was calculated to be less than 0.01%. The initial arc length ranged from 7 mm to 10 mm in the present

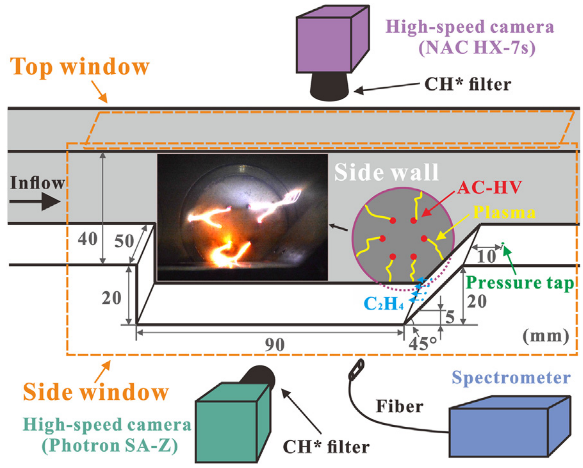

Fig. 1. Schematic of the experimental setup and optical diagnostic system, along with a typical MCGA photo with an exposure time of 1/2500s.

work. The voltage (V) was measured by an HV probe (P6015A, Tektronix) with an uncertainty of ±3%. The current (I) was measured by a current probe (6600, Pearson) with an uncertainty of ±1%. The voltage and current waveforms were recorded by an oscilloscope (DP04104, Tektronix).

2.3. Diagnostics

The optical diagnostic system consisted of high-speed photography and optical emission spectroscopy. Two high-speed cameras equipped with CH* filters (430 ± 10 nm) were arranged at the side view (SA-Z, Photron) and the top view (Memrecam HX-7s, NAC), which were used to record CH* chemiluminescence from the flame simultaneously. The frame rate and the exposure time were set as 10,000 Hz and 99 μs, respectively. An optical fiber spectrometer (Ocean Optics^{TM}, 200–800 nm) was used to capture the emission spectra of the MCGA plasma and the flame. The spectral resolution and collection time of the spectrometer was 0.1 nm and 10 ms, respectively. The collected spectra were correlated to the NIST database. A digital delay/pulse generator (DG535, Stanford Research System) synchronized the cameras and spectrometer with the direct-connect facility.

3. Results and discussion

3.1. Enhancement of flame BL by MCGA plasma

Six cases of the experiments with different FFRs are listed in Table 1. The flame BL can always be defined as the condition for which excessive or insufficient fuel is injected, causing the flame to be blown out. Thus, in the present work, the flame BL

Table 1
Experimental arrangements of all tests.

Case	Fuel flow rate (g/s)	Flame state
1	7.5	Self-sustained
2	7.7	Plasma-sustained
3	8.6	Plasma-sustained
4	9.7	Plasma-sustained
5	10.4	Partially blown out
6	11.8	Completely blown out

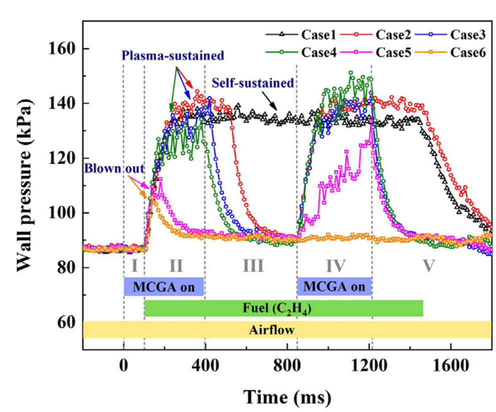

Fig. 2. Wall static pressure evolutions with the same time sequence for Cases 1–6.

can be defined by the FFR. Wall pressures with the same time sequence in the six cases are shown in Fig. 2. The same MCGA plasma was used in the experiment to guarantee the same working conditions. According to the MCGA and fuel operations, the whole combustion process could be divided into Stages I–V. Stage I was regarded as a nature discharge stage, in which the discharge characteristics

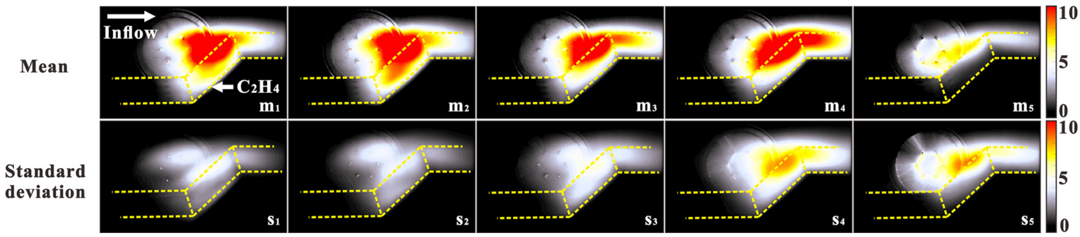

Fig. 3. Mean (m_1–m_5) and standard deviation (s_1–s_5) images for CH* chemiluminescence at Stage IV of Cases 1–5. The images are in the same colormap. Subscripts 1–5 represent Cases 1–5, respectively.

of the MCGA plasma in the non-reacting supersonic flow were recorded. After the C_2H_4 was supplied at Stage II, the MCGA plasma could directly ignite the scramjet combustor. After a rest time of Stage III, the MCGA was turned on again at Stage IV. Stages II–IV were used to test the ability of the MCGA plasma to expand the flame BL in the scramjet combustor. If the MCGA was turned off, the flame would be hardly sustained and blown out again in Stage V.

In Case 1, with FFR = 7.5 g/s, the wall pressure increased to ~135 kPa at the beginning of Stage II, and it was almost kept the same until the first half of Stage V, which meant the flame could be self-sustained during the whole fueling duration. However, the MCGA plasma worked at Stages II and IV, which had no apparent effects on the flame with FFR = 7.5 g/s.

The MCGA plasma was effective in enhancing the flame in Case 2. After the MCGA was off at 400 ms, the wall pressure was only kept for a short time of 150 ms, which meant the cavity-based flame could not be self-sustained for a long time at this FFR. Once the plasma was turned off, the wall pressure gradually decreased to 90 kPa, which meant that the flame was blown out due to the highly unsteady cavity recirculation flow and the local fuel-rich environment. If the MCGA was turned on again after being blown out, the flame was directly ignited in Stage IV. It should be found that the flame could not be self-sustained at FFRs longer than 7.5 g/s in the absence of the MCGA plasma. Consequently, based on the experimental conditions in the present work, an estimate of the flame BL was given that FFR = 7.5 g/s.

Once the FFR was greater than the flame BL, the flame tended to be blown out without the MCGA plasma. However, the flame could be sustained in the presence of the MCGA plasma in Cases 2–4, which was clearly shown in the curves of the wall pressure in Fig. 2. It should be noted that the wall pressure occasionally fluctuated, especially in Case 4. The fluctuations meant that the flame enhancement by the MCGA plasma struck a dynamic balance with the flame blowout by the supersonic flow. This FFR (9.7 g/s) was regarded as the flame BL in the presence of the MCGA plasma, which suggested that the flame BL could be extended by 29% in the presence of the plasma.

When the FFR continued to grow in Cases 5 and 6, the wall pressures of Stages II and IV became relatively low in the presence of the MCGA plasma, sometimes almost equal to the pressure of the non-reacting flow. This finding suggests the flame was partially and completely blown out in Cases 5 and 6, respectively, showing that the flame could not be sustained in the presence of the MCGA plasma at those FFRs.

The flame mean (m_1–m_5) and standard deviation (s_1–s_5) images for CH* chemiluminescence at Stage IV of Cases 1–5 are shown in Fig. 3, respectively. The mean and standard deviation images use the legend ranging from 0~10 to represent the relative CH* emission intensity. Four sonic fuel injectors were parallel to the cavity floor, and the fuel was mixed with the airflow at the recirculation zone near the aft ramp of the cavity. Thus, the flame mainly resided in the low-speed recirculation zone, in which the heat release increased the wall pressures, as shown in Fig. 2. The mean images in Fig. 3 show that the flame almost covered the bottom of the tunnel in m_1–m_4. However, the flame in m_5 existed near the sidewall instead of covering the bottom of the tunnel. The reason is that the local environment in Case 5 was too rich to stabilize the flame completely, so only a part of the flame was observed in m_5 of Fig. 3. The results agree with those shown in Fig. 2.

The standard deviation images for CH* chemiluminescence represent the degree of flame stability. Fig. 2 shows that the flame in Case 1 was self-sustained whether the MCGA plasma was added or not. However, the flame in Cases 2–5 could only be sustained in the presence of the MCGA plasma. The same trend was obtained in Fig.3. A slight standard deviation was seen in s_1–s_3 of Fig. 3, indicating that the flame could be well sustained by the MCGA plasma. A relatively large standard deviation in s_4 denoted that the flame fluctuations in Case 4 became more substantial than those in Cases 1–3. The relatively large standard deviation in s_5 denoted the large flame fluctuations in Case 5 due to the flame hardly being sustained.

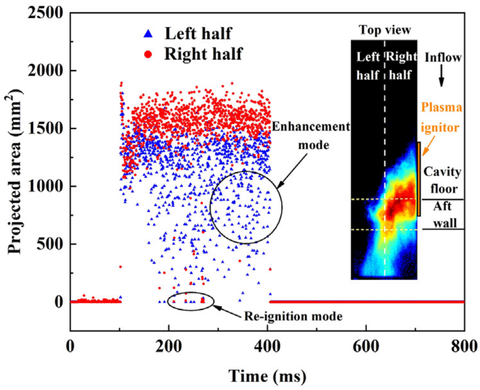

Fig. 4. Time-resolved CH* chemiluminescence projected areas from the top view of the tunnel. The blue and red dots represent the time-resolved projected areas at the left half and right half, respectively. The typical enhancement mode and re-ignition mode are circled. The illustration is a typical image of enhancement mode.

From the whole combustion process corresponding to the wall pressure evolution and the CH* chemiluminescence images, it can be inferred that the MCGA plasma played an essential role in improving the flame BL. As the FFR increased in different cases, the MCGA plasma can extend the flame BL by 29%.

3.2. *Modes of plasma-enhanced bl*

To research the modes of the plasma-enhanced BL, projected flame areas obtained from CH* chemiluminescence images in Case 4 are shown in Fig. 4. The time-resolved (0.4 ms in time step) images on the top view (37 mm in width) were divided into two parts (left half and right half) from the middle plane of the combustor. The MCGA plasma was arranged on the right sidewall of the combustor. The fuel needed a period of time (~tens of milliseconds) to establish a relatively stable FFR. At this period, the FFR gradually increased from 0 g/s to the set value of 9.7 g/s, and the flame was stable until the FFR was beyond the flame BL. Fig. 4 shows that the projected flame areas from the left half and right half were almost the same at the beginning of Stage II.

After the FFR reached 7.9 g/s, flame areas from the right half were always greater than that from the left half for all the stages. A typical CH* chemiluminescence image is shown in an insert of Fig. 4, in which the right half and the left half are labeled. Fig. 4 shows that the flames located at the left half without directly attaching the MCGA plasma were weaker or more frequently blowout than those at the right half enhanced directly by the MCGA plasma. Meanwhile, it should be noted that the flame at the right half could always be stabilized by the MCGA plasma, which could be observed from the large projected areas in Fig. 4. Even if the global flame was blown out, the flame located at the right half kept for a long period than those at the left half, and then the global flame could be observed again due to enhancement of the MCGA plasma. According to whether the global flame was completely blown out or not, the modes of the MCGA plasma-enhanced BL could be defined as enhancement mode and re-ignition mode.

The representative enhancement mode and re-ignition mode processes in Case 4 were synchronously recorded by high-speed CH* chemiluminescence images from both the top and side views of the combustor, as shown in Figs. 5 and 6, respectively. The outlines for the cavity and tunnel and the top window are marked with yellow lines and dotted white lines, respectively.

In enhancement mode, due to a large number of reactive species in the flame, the flame emission dominated the MCGA plasma, so the MCGA plasma was invisible in Fig. 5a. Fig. 5 shows that the flame started to weaken from the side (the left half in Fig. 4) far from the MCGA plasma. The flame directly attaching the MCGA plasma was still stabilized. The MCGA plasma ignited the flame to generate flame kernels at 223.4 ms in Fig. 5b. Although the right part of the remained flame was blown out, the flame kernels continuously merged into a large flame. This flame gradually propagated and developed into a global flame.

In Case 4, the re-ignition mode can be identified. When the global flame was occasionally blown out, the MCGA plasma could re-ignite the flame in a short time. In the re-ignition mode process, when the global flame was blown out, the MCGA discharge type could be operated at a spark-type discharge [12] and continuously ignite the C_2H_4/air mixture to generate flame kernels. The intensity and volume of the flame kernels were affected by the plasma characteristics. If the flame kernels had enough species and volume, they were able to induce an initial flame. At 274.1 ms, the initial flame propagated to the expansion section and formed a flame base, which finally developed into a global flame.

The MCGA plasma-assisted combustion processes indicated that the MCGA plasma had benefits in generating and enhancing the flame base due to igniting the unburnt C_2H_4/air mixture far away from the flame front. The flame base was crucial to establishing a global flame due to supplying heat and reactive species continuously. Even if the flame base was blown out, the MCGA plasma could re-ignite the flame base again as soon as possible.

3.3. *Mechanism of plasma-enhanced BL*

To better interpret the above results, the discharge waveforms and emission spectra were measured to compare the differences between the MCGA plasma in a non-reacting supersonic flow

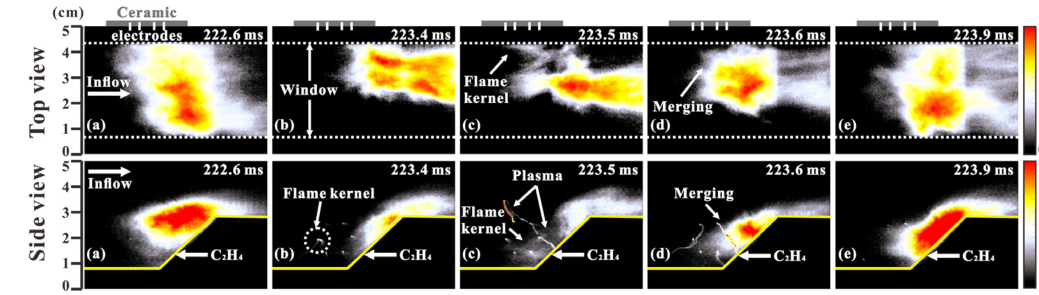

Fig. 5. Typical high-speed CH* chemiluminescence images of the flame enhancement mode process on two views for the MCGA plasma-assisted combustion. The yellow lines and dotted white lines represent the cavity and tunnel outline and the top view window, respectively. The MCGA plasma was arranged on the sidewall.

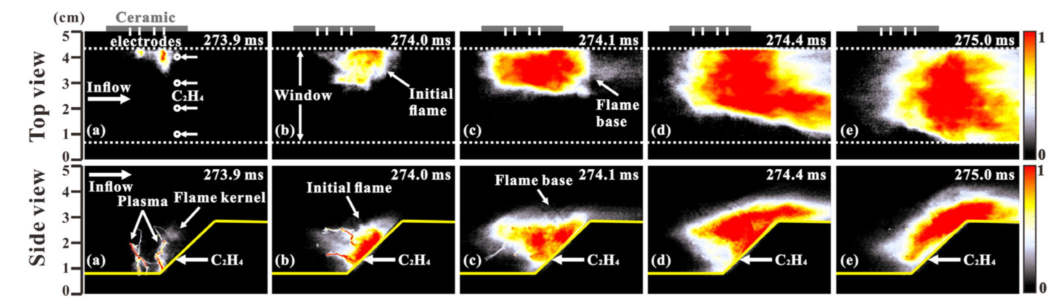

Fig. 6. Typical high-speed CH* chemiluminescence images of the flame re-ignition mode process on two views for the MCGA plasma-assisted combustion. The yellow lines and dotted white lines represent the cavity and tunnel margin and the top view window, respectively. The MCGA plasma was arranged on the sidewall.

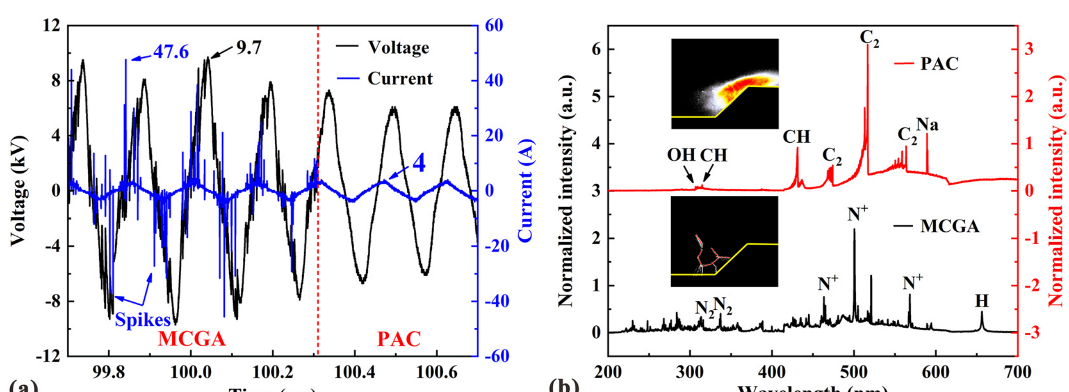

Fig. 7. Characteristics of a) voltage and current, and b) emission spectra for the MCGA stage and the PAC stage, along with two illustrations with an exposure time of 1/10000s.

and a combustible flow for Case 4, as shown in Figs. 7a and 7b, respectively. The discharge process can be divided into two stages by whether the flame emission dominated the MCGA plasma or not: the MCGA stage and the PAC stage. At the MCGA stage, many current spikes were in the current waveform with a peak value of 47.6 A, which indicated that the MCGA discharge type was dominated by the spark-type discharge [12]. The peak voltage and typical current values reached 9.7 kV and 4 A, respectively. The average power of the MCGA plasma at the plasma stage was 5080 W. The spark-to-glow discharge type transitions can be clearly observed in Fig. 7a. The current spikes gradually decreased and soon disappeared in a sub-ms duration in the transition process. At the PAC stage, the MCGA discharge type transferred into a glow-type discharge, which had no current spikes. The aver-

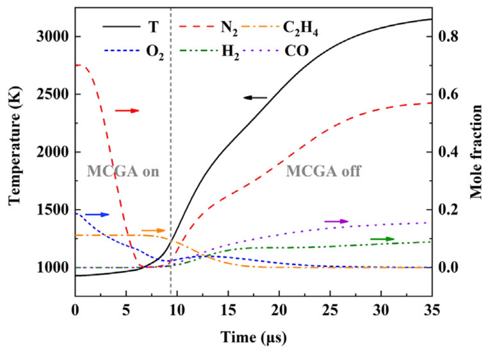

Fig. 8. Temporal evolution of temperature, mole fractions of N_2, C_2H_4, O_2, H_2, and CO by 0-D simulation for Case 4.

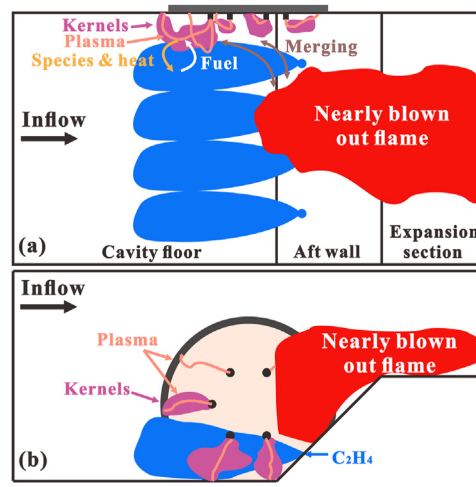

Fig. 9. Sketch of the mechanism for the enhancement of the flame BL by the MCGA plasma: a) top view and b) side view.

age power of the MCGA plasma at the PAC stage was about 3140 W. Combustion in flame propagation mode consumed fuel at flame fronts. Considering the local volume covered by the MCGA plasma and the fuel consumption, the local thermal power by fuel combustion was less than 100 W. The thermal effect can be dominant in the MCGA plasma in local reaction volume.

The spectra provided a qualitative view of the excited species produced by the MCGA stage and the PAC stage. In the spark-type discharge, the MCGA plasma was dominated by atomic spectral lines of N^+ and a few molecular spectral lines of N_2 [19]. When the flame emission dominated the MCGA plasma, discharge type transferred to the glow-type discharge. Typical combustion species (CH and C_2) of C_2H_4 flame dominated the PAC spectra. The Na spectral line probably came from the vitiated airflow used in the air heater. No considerable emission from N^+ and N_2 was detected. The peak intensity ratio of OH^*(310 nm)/CH^*(431 nm) could be used to determine fuel concentration [20]. The local equivalence ratio (φ) was roughly estimated to be ~1.8 by an extra calibration experiment using a plat premixed McKenna burner.

Temporal evolution of temperature, mole fractions of N_2, C_2H_4, O_2, H_2, and CO by 0-D simulation using plasma kinetics solver (ZDPlasKin) and chemical reaction solver (Chemkin III) [15] for Case 4 is shown in Fig. 8. The residence time of the mixture interacting with the MCGA plasma was ~9.3 μs. In the presence of the MCGA plasma, a large amount of N_2 was consumed, generating N_2^*, N_2^+, and N^+. The similar experiment result was shown in the emission spectra. The C_2H_4 gradually participated in the chemical reaction, leading to the increasing heat release and temperature. The ignition delay can be defined as the highest rate of temperature rise. From the simulation results, the ignition delay of the C_2H_4/air mixture was 10.3 μs. As the chemical reactions continued, the mole fraction of H_2 and CO began to rise. Accumulation of H_2 and CO was helpful in promoting successful ignition.

The PAC processes in Figs. 5 and 6 show that the MCGA plasma had different effects on combustion, which were distinguished by whether the flame emission dominated the MCGA plasma or not. If the local flame had been established or the flame emission dominated the MCGA plasma, the MCGA plasma had a minor thermal effect on the flame, because the heat released by combustion was much more than the MCGA plasma input power [21]. In this case, the MCGA discharge type was operated at the glow-type. Inversely, if the local flame was blown out due to a fuel-rich environment, the re-ignition events by the MCGA plasma were benefited by coupled thermal and kinetic effects, and both the heat and reactive species from the MCGA plasma contributed to the combustion enhancement.

In general, although the PAC processes were divided into the enhancement mode and re-ignition mode, the PAC mechanism was the same. A mechanism sketch for the enhancement of the flame BL on the top and side views is shown in Fig. 9. When a local fuel-rich environment resulted in local flame-outs, the global flame was nearly blown out. The MCGA discharge type transferred to the spark-type discharge due to the flame moving backward. The MCGA plasma showed the thermal and kinetic effects on the local ignition. As more C_2H_4 was ignited by the MCGA plasma, the flame kernels merged to form the flame base and gradually enabled the flame base to develop into a global flame. Moreover, if the flame base was completely blown out, the MCGA plasma with the spark-type discharge also showed excellent thermal and kinetic effects on re-ignition, especially in a non-reacting flow. The coupled products of high-temperature and reactive species (N^+) from the MCGA plasma were favorable for igniting C_2H_4 to form the flame kernels. If a suitable condition (e.g., local equivalence ratio, velocity) appeared at the local position in the high-turbulence supersonic flow, the flame kernels developed into an initial flame and finally established a global flame.

When the FFR was under the flame BL, local flameouts could not affect the existence of the flame base. The MCGA plasma directly assisted the flame base through the enhancement mode. When the FFR was close to the flame BL, the flame base was occasionally blown out. The MCGA plasma was operated either in the enhancement mode or the re-ignition mode. When the FFR was beyond the flame BL, extinction can be frequently observed, and the re-ignition mode dominated the PAC process. In this case, the MCGA plasma gradually decreased the effects on the enhancement of the flame BL as the FFR increasing.

4. Conclusions

In conclusion, this work shows that the multichannel gliding arc (MCGA) plasma can effectively enhance the flame blowout limit (BL) in a scramjet combustor. The enhancement of the flame BL in the C_2H_4-fueled and cavity-based model scramjet combustor by the MCGA plasma was investigated using CH* chemiluminescence imaging, wall static pressure measurement, optical emission spectroscopy, and discharge waveform measurements.

The supersonic flow rate was kept constant for these experiments, and the fuel flow rate (FFR) was slowly increased, enabling the flame to change from a self-sustained state to a blowout state. The MCGA plasma was used to enhance the flame BL. For the current configuration and fueling condition, the flame BL effectively increased from FFR = 7.5 g/s to the FFR = 9.7 g/s in the presence of the MCGA plasma, showing that the flame BL had an increase of ~29% in the presence of the MCGA plasma. If the FFR is larger than 10.4 g/s, the flame could not be sustained for a long time in the presence of the MCGA plasma.

Two plasma-assisted combustion (PAC) modes, including enhancement mode and re-ignition mode, contributed to the extension of the flame BL. Although the two modes showed different processes, their PAC mechanisms were the same. The spark-type discharge of the MCGA plasma showed substantial thermal and kinetic effects on the ignition of the unburnt fuel/air mixture. The exchanges of species and heat generated by the MCGA plasma in the fuel/air mixture processes were favorable for the ignition events. The dynamic balance between the ignition by the MCGA plasma and the flame blowout by the supersonic flow was essential for the enhancement of the flame BL in scramjet combustors.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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